

CHUQUR O'RGANISH USULLARI · Ma'ruza konspekti (Lecture Note)

1-ma'ruza. Sun'iy neyron va perceptron

2026–2027 o'quv yili

Reja. 1) Sun'iy neyronning matematik modeli va geometriyasi. 2) Faollashtirish funksiyalari (sigmoid, tanh, ReLU, Leaky ReLU) va ularning hosilalari. 3) Perceptron o'rganish qoidasi va yaqinlashish teoremasi. 4) XOR masalasi va yashirin qatlam. 5) Logistik neyron, binary cross-entropy va gradient tushish. 6) Universal approksimatsiya. 7) Python (NumPy, PyTorch). 8) Nazorat savollari va mashqlar.

O'quv natijalari. Ma'ruzadan so'ng talaba: neyron chiqishini qo'lda va dasturda hisoblaydi; faollashtirish funksiyalari va hosilalarining xossalari isbotlaydi; perceptron algoritmini bajaradi va uning yaqinlashish shartini asoslaydi; XOR ning chiziqli ajralmasligini isbotlaydi va uni yashirin qatlam bilan yechadi; binary cross-entropy gradientini chiqaradi va autograd bilan tekshiradi.

Tayanch iboralar. sun'iy neyron, og'irlik, siljish (bias), oldindan faollashtirish (pre-activation), faollashtirish funksiyasi, giperyuza, chiziqli ajraluvchanlik, perceptron, margin, XOR, yashirin qatlam, logit, binary cross-entropy, gradientning so'nishi (vanishing gradient), universal approksimatsiya.

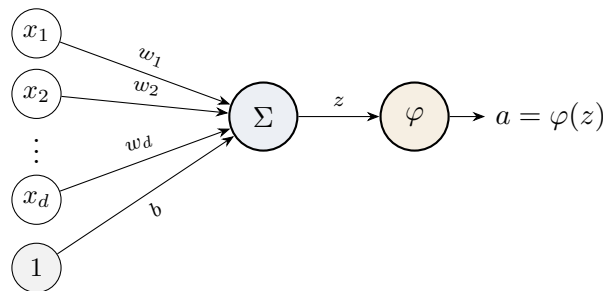
1. Sun'iy neyronning matematik modeli

Chuqur o'rganishdagi har qanday tarmoq — eng sodda hisoblash birligi, *sun'iy neyronning* ko'p marta takrorlanishi va ulanishidir. Shu sababli kurs butun tarmoqdan emas, bitta neyron boshlanadi: uning nimani qila olishi va nimani qila olmasligi keyingi barcha arxitekturalarning sababini tushuntiradi.

Ta'rif 1.1 (Sun'iy neyron). Kirish vektori $\mathbf{x} \in \mathbb{R}^d$, og'irliklar $\mathbf{w} \in \mathbb{R}^d$, siljish $b \in \mathbb{R}$ va faollashtirish funksiyasi $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ berilgan bo'lsin. Sun'iy neyron quyidagi ikki qadamli akslantirishdir:

$$z = \mathbf{w}^T \mathbf{x} + b = \sum_{i=1}^d w_i x_i + b, \quad a = \varphi(z).$$

z — oldindan faollashtirish (pre-activation, chiziqli qism), a — neyron chiqishi (activation). Neyronning o'rganiladigan parametrlari $\theta = (\mathbf{w}, b)$ — jami $d + 1$ ta son.



Bias trick. $\tilde{\mathbf{x}} = (\mathbf{x}; 1) \in \mathbb{R}^{d+1}$, $\tilde{\mathbf{w}} = (\mathbf{w}; b)$ belgilansa, $z = \tilde{\mathbf{w}}^T \tilde{\mathbf{x}}$. Nazariy isbotlarda siljishni shu tarzda og'irliklar ichiga “yashirish” qulay.

1.1. Geometrik ma'no

$\mathbf{w} \neq \mathbf{0}$ bo'lsa, $\{\mathbf{x} : \mathbf{w}^T \mathbf{x} + b = 0\}$ to'plam $\mathbb{R}^d$ dagi giperyuza ($d = 2$ da to'g'ri chiziq, $d = 3$ da tekislik). U fazoni ikki yarim fazoga ajratadi: $z > 0$ va $z < 0$.

Xossa 1.2. $\mathbf{w}$ — giperyuzaga normal vektor, $\mathbf{x}_0$ nuqtadan giperyuzagacha masofa $\rho = \frac{|\mathbf{w}^T \mathbf{x}_0 + b|}{\|\mathbf{w}\|}$.

Demak $|z|$ — masofaga proporsional “ishonch” o'lchovi.

Isbot. Giperyuzadagi ikki nuqta uchun $\mathbf{w}^T (\mathbf{x}_1 - \mathbf{x}_2) = 0$, ya'ni $\mathbf{w}$ har qanday yo'naltiruvchi vektorga ortogonal. $\mathbf{x}_0$ ning proyeksiyasi $\mathbf{x}_0 - t\mathbf{w}$ ko'rinishda; $\mathbf{w}^T (\mathbf{x}_0 - t\mathbf{w}) + b = 0$ dan $t = (\mathbf{w}^T \mathbf{x}_0 + b) / \|\mathbf{w}\|^2$, masofa esa $|t| \|\mathbf{w}\| = |\mathbf{w}^T \mathbf{x}_0 + b| / \|\mathbf{w}\|$. □

Natijada, qadamsimon φ bilan bitta neyron faqat *chiziqli ajraluvchi* sinflarni ajrata oladi.

Ta'rif 1.3. $\{(\mathbf{x}^{(k)}, y_k)\}_{k=1}^n$, $y_k \in \{0, 1\}$ to'plam *chiziqli ajraluvchi* deyiladi, agar shunday $\mathbf{w}, b$ topilsaki, $y_k = 1$ da $\mathbf{w}^T \mathbf{x}^{(k)} + b > 0$ va $y_k = 0$ da $\mathbf{w}^T \mathbf{x}^{(k)} + b < 0$ bo'lsa.

Xossa 1.4 (Chiziqliksiz chuqurlik foydasiz). Agar $\varphi(z) = z$ bo'lsa, ixtiyoriy sondagi ketma-ket chiziqli qatlamlar bitta chiziqli qatlamga teng.

Isbot. $W_2(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 = (W_2W_1)\mathbf{x} + (W_2\mathbf{b}_1 + \mathbf{b}_2) = W\mathbf{x} + \mathbf{b}$. Induksiya bo'yicha L qatlam uchun ham affin akslantirishlar kompozitsiyasi affin. $\square$

Neyronlar qatlami. Bir xil kirishni qabul qiluvchi m ta neyron *to'liq bog'langan qatlam* (fully connected, `nn.Linear`) hosil qiladi. Paket $X \in \mathbb{R}^{n \times d}$ (qatorlar — namunalar) uchun

$$Z = XW + \mathbf{1}b^T \in \mathbb{R}^{n \times m}, \quad A = \varphi(Z) \text{ (elementlar bo'yicha),}$$

$W \in \mathbb{R}^{d \times m}$ ning j -ustuni — j -neyron og'irliklari; parametrlar soni $m(d+1)$. Masalan, 28×28 tasvir ($d = 784$) va $m = 128$: $128 \cdot 785 = 100\,480$ parametr. Barcha namunalar va neyronlar uchun hisob bitta matritsa ko'paytmasi — GPU aynan shuni tez bajaradi.

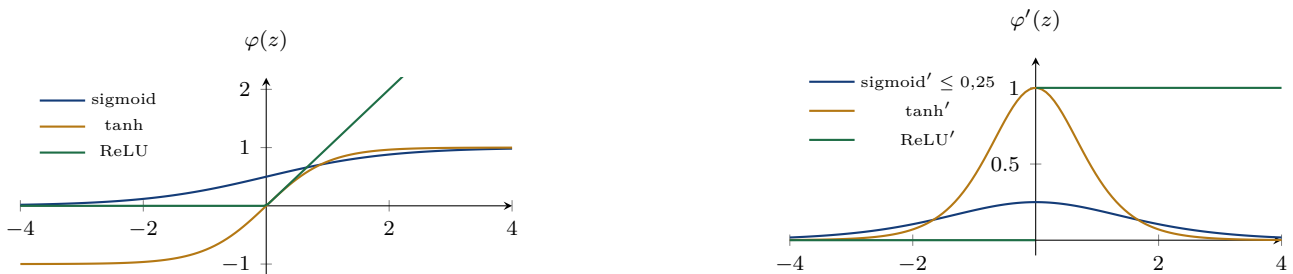
Amaliy talqin: neyron — tarozi

Tibbiy skrining: x_1 — harorat, x_2 — qon bosimi. Neyron har bir belgiga “ahamiyat” w_i beradi, b esa qaror ostonasini suradi; $z > 0$ bo'lsa — “xavf guruhi”. $|w_i|$ ning kattaligi belgining ta'sirini, ishorasi — yo'nalishini bildiradi (belgilar bir xil masshtabga keltirilgan bo'lsa). Kredit skoringi, spam filtri, sifat nazorati — barchasi shu sxemaning xususiy holi.

2. Faollashtirish funksiyalari

Nomi	$\varphi(z)$	$\varphi'(z)$	Qiymatlar sohasi, qo'llanishi
sigmoid	$\sigma(z) = \frac{1}{1 + e^{-z}}$	$\sigma(z)(1 - \sigma(z))$	$(0, 1)$; ikkilik chiqish (ehtimollik)
tanh	$\frac{e^z - e^{-z}}{e^z + e^{-z}}$	$1 - \tanh^2 z$	$(-1, 1)$; nolga nisbatan simmetrik
ReLU	$\max(0, z)$	$\mathbf{1}[z > 0]$	$[0, \infty)$; yashirin qatlamlar standarti
Leaky ReLU	$\max(\alpha z, z)$, $0 < \alpha < 1$	$\mathbf{1}(z > 0)$, $\alpha(z < 0)$	$\mathbb{R}$; “o'lik” ReLU ga qarshi
softplus	$\ln(1 + e^z)$	$\sigma(z)$	$(0, \infty)$; ReLU ning silliq varianti

ReLU $z = 0$ da differensiallanmaydi; kutubxonalarda shartli ravishda $\text{ReLU}'(0) = 0$ olinadi (subgradient). Zamonaviy modellarda GELU $z\Phi(z)$ (Φ — standart normal taqsimot funksiyasi) ham keng ishlatiladi.



Teorema 2.1. Ixtiyoriy $z \in \mathbb{R}$ uchun: (i) $\sigma'(z) = \sigma(z)(1 - \sigma(z))$; (ii) $\sigma(-z) = 1 - \sigma(z)$; (iii) $0 < \sigma'(z) \leq \frac{1}{4}$, tenglik faqat $z = 0$ da; (iv) $\tanh z = 2\sigma(2z) - 1$, demak $\tanh'(0) = 1$.

Isbot. (i) $\sigma'(z) = \frac{e^{-z}}{(1 + e^{-z})^2} = \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} = \sigma(z)(1 - \sigma(z))$. (ii) $1 - \sigma(z) = \frac{e^{-z}}{1 + e^{-z}} = \frac{1}{e^z + 1} = \sigma(-z)$. (iii) $s = \sigma(z) \in (0, 1)$ uchun $s(1 - s) = \frac{1}{4} - (s - \frac{1}{2})^2 \leq \frac{1}{4}$, tenglik $s = \frac{1}{2} \iff z = 0$. (iv) $2\sigma(2z) - 1 = \frac{1 - e^{-2z}}{1 + e^{-2z}} = \frac{e^z - e^{-z}}{e^z + e^{-z}}$; zanjir qoidasi bo'yicha $\tanh'(0) = 4\sigma'(0) = 1$. $\square$

Natija 2.2 (Gradientning so'nishi). Har bir qatlamda gradient φ' ga ko'paytiriladi (2-ma'ruza). Sigmoid uchun bu ko'paytuvchi $\leq 0,25$, shuning uchun L qatlamdan keyin faollashtirish hissasi $\leq 0,25^L$: $L = 5$ da $9,8 \cdot 10^{-4}$, $L = 20$ da $9,1 \cdot 10^{-13}$. To'yingan sohada holat yanada yomon: $\sigma'(6) \approx 0,00247$. ReLU uchun musbat sohada $\varphi' = 1$ — signal susaymaydi; evaziga $z < 0$ da gradient nol ("o'lik ReLU"), buni Leaky ReLU ($\alpha = 0,01$) yumshatadi.

Qaysi joyda qaysi funksiya

Yashirin qatlamlar — ReLU (yoki Leaky ReLU, GELU); ikkilik chiqish — sigmoid; ko'p sinfli chiqish — softmax (2-ma'ruza); regressiya chiqishi — faollashtirishsiz (chiziqli). Yashirin qatlamda sigmoid — odatda eskirgan tanlov.

3. Perceptron: birinchi o'rganish qoidasi

Tarixiy lavha. McCulloch va Pitts (1943) mantiqiy neyron modelini, Rozenblatt (1958) o'rganuvchi perceptronni taklif qildi. Minsky va Papert (1969) bitta qatlamli perceptron XOR ni hisoblay olmasligini ko'rsatgach, qiziqish susaydi. Ko'p qatlamli tarmoqlarni backpropagation bilan o'qitish (Rumelhart, Hinton, Williams, 1986) va keyinchalik GPU hamda katta ma'lumotlar (AlexNet, 2012) chuqur o'rganishni qayta tikladi.

Rozenblatt perceptroni (1958) qadamsimon chiqishli neyron: $\hat{y} = \mathbf{1}[w^T x + b \geq 0]$. G'oya: xato qilinsa, parametrlarni shu namunani to'g'ri tomonga suradigan yo'nalishda o'zgartirish.

Algoritm 1. Perceptron (Rozenblatt qoidasi)

Kirish: $\{(x^{(k)}, y_k)\}_{k=1}^n$, $y_k \in \{0, 1\}$; qadam $\eta > 0$; epoxalar chegarasi T .

1. $w \leftarrow \mathbf{0}$, $b \leftarrow 0$.
2. $t = 1, \dots, T$ uchun: $xato \leftarrow 0$; har bir $k = 1, \dots, n$ uchun
 - (a) $\hat{y} \leftarrow \mathbf{1}[w^T x^{(k)} + b \geq 0]$, $\delta \leftarrow y_k - \hat{y} \in \{-1, 0, 1\}$;
 - (b) agar $\delta \neq 0$: $w \leftarrow w + \eta \delta x^{(k)}$, $b \leftarrow b + \eta \delta$, $xato \leftarrow xato + 1$.
3. Agar $xato = 0$ bo'lsa — to'xtash (barcha namunalar to'g'ri).

Chiqish: w, b . T cheklovi majburiy: ajralmaydigan to'plamda algoritm to'xtamaydi.

Qoida nega ishlaydi: $\delta = +1$ ($y = 1$, $\hat{y} = 0$) bo'lsa, yangilangandan keyin z qiymati $\eta(\|x\|^2 + 1) > 0$ ga ortadi; $\delta = -1$ da shuncha kamayadi. $w_0 = \mathbf{0}$ bo'lganda η faqat parametrlarni masshtablaydi va bashoratlarga ta'sir qilmaydi, shuning uchun misolda $\eta = 1$ olamiz (butun sonli, aniq arifmetika).

Misol 3.1 (AND funksiyasi). $x \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, $y = (0, 0, 0, 1)$, $\eta = 1$, namunalar shu tartibda.

Yechim. 1-epoxa. (0,0): $z = 0 \Rightarrow \hat{y} = 1 \neq 0$, $\delta = -1$: $w = (0, 0)$, $b = -1$. (0,1): $z = -1$, $\hat{y} = 0$ — to'g'ri. (1,0): $z = -1$ — to'g'ri. (1,1): $z = -1 \Rightarrow \hat{y} = 0 \neq 1$, $\delta = +1$: $w = (1, 1)$, $b = 0$. Xatolar: 2. Keyingi epoxalar oxiridagi holat:

Epoxa	1	2	3	4	5	6
Xatolar soni	2	3	3	2	1	0
w	(1, 1)	(2, 1)	(2, 1)	(2, 2)	(2, 1)	(2, 1)
b	0	-1	-2	-2	-3	-3

6-epoxada xato yo'q: $z = (-3, -2, -1, 0)$, $\hat{y} = (0, 0, 0, 1)$ — AND. Jami 11 ta yangilanish. E'tibor bering: (1,1) nuqta aynan chegarada ($z = 0$) yotadi va faqat $\geq$ shart tufayli to'g'ri sinflanadi — margin nolga teng, bunday yechim shovqinga chidamsiz. $\triangleleft$

Izoh 3.2. $\eta = 0,2$ da aniq arifmetikada natija shunchaki 0,2 marta kichrayadi: $w = (0,4; 0,2)$, $b = -0,6$. Suzuvchi nuqtali hisobda esa 0,2 ikkilik sanoq tizimida aniq ifodalanmagani uchun chegaradagi $z = 0$ holatlarda natija o'zgarishi mumkin (masalan, $-5,55 \cdot 10^{-17} < 0$): NumPy'da algoritm 4-epoxada $b \approx -0,4$ bilan to'xtaydi. Chegaraviy tenglikka tayanmang.

Teorema 3.3 (Novikoff, 1962). Bias trick bilan $\tilde{\mathbf{x}}^{(k)} = (\mathbf{x}^{(k)}; 1)$, $\tilde{y}_k = 2y_k - 1 \in \{-1, 1\}$ bo'lsin, $\|\tilde{\mathbf{x}}^{(k)}\| \leq R$ va shunday birlik vektor $\mathbf{u}$ hamda $\gamma > 0$ mavjud bo'lsinki, barcha k uchun $\tilde{y}_k \mathbf{u}^T \tilde{\mathbf{x}}^{(k)} \geq \gamma$. U holda perceptron (ixtiyoriy tartibda, $\mathbf{w}_0 = \mathbf{0}$) ko'pi bilan $(R/\gamma)^2$ ta yangilanish qiladi.

Isbot. Yangilanish faqat $\tilde{y}_k \tilde{\mathbf{w}}^T \tilde{\mathbf{x}}^{(k)} \leq 0$ bo'lganda bo'ladi va $\delta = \tilde{y}_k$, ya'ni $\tilde{\mathbf{w}} \leftarrow \tilde{\mathbf{w}} + \eta \tilde{y}_k \tilde{\mathbf{x}}^{(k)}$. m ta yangilanishdan keyin: $\mathbf{u}^T \tilde{\mathbf{w}}_m \geq \mathbf{u}^T \tilde{\mathbf{w}}_{m-1} + \eta \gamma \geq m \eta \gamma$; $\|\tilde{\mathbf{w}}_m\|^2 = \|\tilde{\mathbf{w}}_{m-1}\|^2 + 2\eta \tilde{y}_k \tilde{\mathbf{w}}_{m-1}^T \tilde{\mathbf{x}}^{(k)} + \eta^2 \|\tilde{\mathbf{x}}^{(k)}\|^2 \leq \|\tilde{\mathbf{w}}_{m-1}\|^2 + \eta^2 R^2 \leq m \eta^2 R^2$. Koshi–Bunyakovskiy: $m \eta \gamma \leq \mathbf{u}^T \tilde{\mathbf{w}}_m \leq \|\tilde{\mathbf{w}}_m\| \leq \sqrt{m} \eta R$, bundan $m \leq (R/\gamma)^2$. $\square$

Teorema ikki tomonlama saboq beradi: to'plam ajraluvchi bo'lsa algoritmi albatta to'xtaydi, margin γ qancha kichik bo'lsa, shuncha sekin. Ajralmaydigan to'plamda esa perceptron “yechim yo'q” deb xabar bermaydi — cheksiz aylanadi.

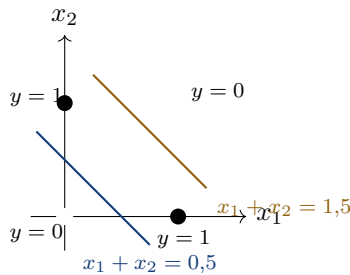
Izoh 3.4. Perceptron qoidasi $\ell(z, \tilde{y}) = \max(0, -\tilde{y}z)$ (“perceptron loss”) uchun stoxastik subgradient tushishga mos keladi: xatoda $\partial \ell / \partial \mathbf{w} = -\tilde{y} \mathbf{x}$, ya'ni $\mathbf{w} \leftarrow \mathbf{w} + \eta \tilde{y} \mathbf{x}$.

4. XOR: bitta neyron yetmaydigan joy

XOR: $(0, 0) \mapsto 0$, $(0, 1) \mapsto 1$, $(1, 0) \mapsto 1$, $(1, 1) \mapsto 0$. Bir sinf nuqtalari kvadratning qarama-qarshi uchlarida joylashgan.

Teorema 4.1 (Minsky–Papert). XOR to'plami chiziqli ajraluvchi emas.

Isbot. Faraz qilaylik, $\mathbf{w}, b$ ajratadi. U holda $b < 0$, $w_2 + b > 0$, $w_1 + b > 0$, $w_1 + w_2 + b < 0$. O'rta ikki tengsizlikni qo'shsak $w_1 + w_2 + 2b > 0$, chetki ikkitasini qo'shsak $w_1 + w_2 + 2b < 0$ — ziddiyat. $\square$



Bitta chiziqli qora nuqtalarni oqlaridan ajrata olmaydi; ikki chiziqli orasidagi yo'lak esa aynan $y = 1$ sinfi. Yashirin qatlam shu ikki chiziqli yaratadi.

NumPy'da perceptron XOR uchun 100 epoxada ham to'xtamaydi va har epoxada kamida bitta xato qoladi (6-bo'lim). Yechim — *yashirin qatlam*: bir necha neyron kirishni yangi koordinatalarga o'tkazadi, bu koordinatalarda sinflar chiziqli ajraladi.

Misol 4.2 (XOR ni qo'lda qurish). $h_1 = \text{ReLU}(x_1 + x_2)$, $h_2 = \text{ReLU}(x_1 + x_2 - 1)$, $\hat{y} = h_1 - 2h_2$.

Yechim.

$\mathbf{x}$	h_1	h_2	$h_1 - 2h_2$
(0, 0)	0	0	0
(0, 1)	1	0	1
(1, 0)	1	0	1
(1, 1)	2	1	0

Matritsa shaklida: $W_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{b}_1 = (0; -1)$, $\mathbf{w}_2 = (1; -2)$, $b_2 = 0$ — 9 parametrli tarmoq XOR ni aniq hisoblaydi. Yashirin qatlam $(0, 1)$ va $(1, 0)$ ni bitta $(1, 0)$ nuqtaga “yopishtirdi”, $(1, 1)$ esa $(2, 1)$ ga o'tdi — yangi fazoda sinflar to'g'ri chiziqli bilan ajraladi. $\triangleleft$

Izoh 4.3. Yashirin neyron bitta bo'lsa ($m = 1$), XOR yechilmaydi: chiqish $\mathbf{x}$ ga yagona skalyar $u = \mathbf{w}_1^T \mathbf{x} + b_1$ orqali *monoton* bog'liq (sigmoid, tanh, ReLU monoton), demak qaror sohasi $\{u \geq c\}$ yoki $\{u \leq c\}$ — yana yarim tekislik.

5. Logistik neyron va gradient tushish

Qadamsimon funksiyaning hosilasi deyarli hamma joyda nol, shuning uchun gradient bilan o'qitish uchun *silliqliq* model olinadi: $p = \sigma(z)$ — “ $y = 1$ ” ehtimolligi, z — *logit*.

Ta'rif 5.1 (Binary cross-entropy). Bitta namuna uchun $\ell(z, y) = -y \ln p - (1 - y) \ln(1 - p)$, $p = \sigma(z)$. Teorema 2.1(ii) dan foydalanib, uni sonli barqaror shaklda yozish mumkin:

$$\ell(z, y) = \ln(1 + e^z) - yz.$$

Haqiqatan, $-\ln \sigma(z) = \ln(1 + e^{-z})$, $-\ln(1 - \sigma(z)) = \ln(1 + e^z)$ va $\ln(1 + e^{-z}) = \ln(1 + e^z) - z$. $\ln(1 + e^z)$ np.logaddexp(0,z) bilan har qanday z da toshib ketmasdan hisoblanadi; to'g'ridan-to'g'ri $\ln \sigma(z)$ esa $z = -800$ da $\ln 0 = -\infty$ beradi.

Teorema 5.2 (Logistik neyron gradienti). $\frac{\partial \ell}{\partial z} = \sigma(z) - y = p - y$, demak $\nabla_{\mathbf{w}} \ell = (p - y)\mathbf{x}$, $\frac{\partial \ell}{\partial b} = p - y$. n ta namuna bo'yicha o'rtacha yo'qotish $L = \frac{1}{n} \sum_k \ell_k$ uchun $\nabla_{\mathbf{w}} L = \frac{1}{n} X^T (\mathbf{p} - \mathbf{y})$, bunda $X \in \mathbb{R}^{n \times d}$.

Isbot. $\frac{d}{dz} \ln(1 + e^z) = \frac{e^z}{1 + e^z} = \sigma(z)$, demak $\partial \ell / \partial z = \sigma(z) - y$. Zanjir qoidasi: $\partial z / \partial w_i = x_i$, $\partial z / \partial b = 1$. O'rtachalash chiziqli amal. $\square$

Gradient “xatoning o'zi” $p - y$ ga proporsional: model ishonch bilan to'g'ri bo'lsa, qadam kichik; ishonch bilan xato bo'lsa, katta. Gradient tushish qadami: $\theta \leftarrow \theta - \eta \nabla_{\theta} L$.

Misol 5.3 (Bitta gradient qadami). $\mathbf{x} = (1; 2)$, $y = 1$, $\mathbf{w} = (0,5; -0,3)$, $b = 0,1$, $\eta = 0,1$.

Yechim. $z = 0,5 - 0,6 + 0,1 = 0$, $p = \sigma(0) = 0,5$, $\ell = \ln 2 \approx 0,6931$. $p - y = -0,5$; $\nabla_{\mathbf{w}} \ell = (-0,5; -1)$, $\partial \ell / \partial b = -0,5$. Yangilash: $\mathbf{w} = (0,55; -0,2)$, $b = 0,15$. Endi $z = 0,55 - 0,4 + 0,15 = 0,3$, $p = \sigma(0,3) \approx 0,5744$, $\ell \approx 0,5544$ — yo'qotish kamaydi. Natijalar PyTorch autograd bilan tekshirilgan (6-bo'lim). $\triangleleft$

Xossa 5.4 (Nega MSE emas?). Sigmoid chiqishda $\ell_{\text{MSE}} = \frac{1}{2}(p - y)^2$ olinsa, $\partial \ell_{\text{MSE}} / \partial z = (p - y)p(1 - p)$ — binary cross-entropy gradientidan $p(1 - p) \leq \frac{1}{4}$ ko'paytuvchiga farq qiladi. Model ishonch bilan *xato* qilganda ($p \rightarrow 0$, $y = 1$) bu ko'paytuvchi nolga intiladi va o'rganish to'xtaydi.

Masalan, $z = -5$, $y = 1$: $p \approx 0,00669$; binary cross-entropy gradienti $\approx -0,9933$, MSE gradienti esa $\approx -0,0066$ — taxminan 150 marta kichik. Bundan tashqari $\ell_{\text{MSE}}(\sigma(\tilde{\mathbf{w}}^T \tilde{\mathbf{x}}))$ parametrlar bo'yicha qavariq emas.

5.1. Binary cross-entropy qayerdan keladi va uning qavariqligi

Model $P(y = 1 | \mathbf{x}) = p = \sigma(z)$ ni beradi, ya'ni $y \sim \text{Bernoulli}(p)$ va $P(y | \mathbf{x}) = p^y(1 - p)^{1-y}$. Namunalar mustaqil bo'lsa, haqiqatga o'xshashlik funksiyasi $\prod_k p_k^{y_k}(1 - p_k)^{1-y_k}$; uning manfiy logarifmining o'rtachasi aynan $L = \frac{1}{n} \sum_k \ell(z_k, y_k)$. Demak binary cross-entropy ni minimallashtirish — *maksimal haqiqatga o'xshashlik* (MLE) bahosi.

Teorema 5.5. Logistik neyron ($z = \tilde{\mathbf{w}}^T \tilde{\mathbf{x}}$) uchun $L(\tilde{\mathbf{w}})$ qavariq: $\nabla^2 L = \frac{1}{n} \tilde{X}^T S \tilde{X} \succeq 0$, $S = \text{diag}(p_k(1 - p_k))$.

Isbot. $\nabla L = \frac{1}{n} \tilde{X}^T (\mathbf{p} - \mathbf{y})$ va $\partial p_k / \partial \tilde{\mathbf{w}} = p_k(1 - p_k) \tilde{\mathbf{x}}^{(k)}$, demak $\nabla^2 L = \frac{1}{n} \sum_k p_k(1 - p_k) \tilde{\mathbf{x}}^{(k)} \tilde{\mathbf{x}}^{(k)T}$. Ixtiyoriy $\mathbf{v}$ uchun $\mathbf{v}^T \nabla^2 L \mathbf{v} = \frac{1}{n} \sum_k p_k(1 - p_k) (\mathbf{v}^T \tilde{\mathbf{x}}^{(k)})^2 \geq 0$. $\square$

Bitta neyronda har qanday lokal minimum global. Yashirin qatlamli tarmoqda esa bu yo'qoladi: yashirin neyronlarning o'rnini almashtirish (permutatsiya) yo'qotishni o'zgartirmaydi, shuning uchun har bir minimumning $m!$ tagacha nusxasi bor, ularning o'rtachasi esa odatda minimum emas — yo'qotish qavariq bo'lolmaydi. Quyida (6-bo'lim) buning amaliy oqibatini ko'ramiz.

Xossa 5.6 (Logit — log-odds). $p = \sigma(z)$ bo'lsa, $z = \ln \frac{p}{1 - p}$. Shuning uchun x_i bir birlikka oshsa (qolganlari o'zgarmasa), *odds* $p/(1 - p)$ aniq e^{w_i} marta ko'payadi.

Isbot. $p = 1/(1 + e^{-z}) \Rightarrow e^{-z} = (1 - p)/p \Rightarrow z = \ln \frac{p}{1 - p}$; z ning x_i bo'yicha o'sishi w_i , odds esa e^z . $\square$

Amaliy talqin: kredit skoringi

$z = -1 + 0,6x_1 - 2x_2$, x_1 — oylik daromad (mln so'm), x_2 — qarz/daromad nisbati, p — kreditni o'z vaqtida qaytarish ehtimolligi. Mijoz $\mathbf{x} = (5; 0,4)$: $z = 1,2$, $p = \sigma(1,2) \approx 0,769$, odds = $e^{1,2} \approx 3,32$. Daromad 1 mln ga oshsa, odds $e^{0,6} \approx 1,82$ marta ortadi ($p \approx 0,858$); x_2 0,1 ga oshsa, odds $e^{-0,2} \approx 0,82$ marta kamayadi ($p \approx 0,731$). Bank ostona $p \geq 0,7$ ni tanlasa, qaror chegarasi $z \geq \ln(0,7/0,3) \approx 0,847$ — yana to'g'ri chiziq: ostona faqat b ni suradi.

5.2. Gradient tushishning yaqinlashishi va qadam tanlash

Teorema 5.7 (Tushish lemmasi). f differensiallanuvchi va ∇f Lipshtits-uzluksiz bo'lsin: $\|\nabla f(\mathbf{u}) - \nabla f(\mathbf{v})\| \leq \beta \|\mathbf{u} - \mathbf{v}\|$. U holda $\theta^+ = \theta - \eta \nabla f(\theta)$ uchun $f(\theta^+) \leq f(\theta) - \eta(1 - \frac{\eta\beta}{2}) \|\nabla f(\theta)\|^2$. Xususan, $0 < \eta < 2/\beta$ da har qadamda f kamayadi, $\eta = 1/\beta$ da kamayish kamida $\frac{1}{2\beta} \|\nabla f\|^2$.

Isbot. $f(\mathbf{v}) \leq f(\mathbf{u}) + \nabla f(\mathbf{u})^T(\mathbf{v} - \mathbf{u}) + \frac{\beta}{2} \|\mathbf{v} - \mathbf{u}\|^2$ (Nyuton–Leybnits formulasi va Lipshtits sharti). $\mathbf{u} = \theta$, $\mathbf{v} = \theta^+$ qo'yamiz: $\mathbf{v} - \mathbf{u} = -\eta \nabla f$, bundan talab qilingan tengsizlik. $\square$

Logistik neyron uchun $\nabla^2 L = \frac{1}{n} \tilde{X}^T S \tilde{X} \preceq \frac{1}{4n} \tilde{X}^T \tilde{X}$ (chunki $p(1-p) \leq \frac{1}{4}$), demak $\beta = \lambda_{\max}(\tilde{X}^T \tilde{X})/(4n)$ va $\eta < 2/\beta$ xavfsiz. Juda katta η tebranish yoki uzoqlashishga, juda kichik η sekin o'qitishga olib keladi; neyron tarmoqlarda β noma'lum, shuning uchun η tajriba yo'li bilan (masalan, $10^{-4} \dots 10^0$ oralig'ida) logarifmik to'r bilan) tanlanadi.

Algoritm 2. Mini-batch gradient tushish (SGD)

Kirish: ma'lumot $\{(\mathbf{x}^{(k)}, y_k)\}_{k=1}^n$, paket hajmi B , qadam η , epoxalar E .

1. Parametrlarni tasodifiy (kichik) qiymatlar bilan boshlash.
 2. Har bir epoxada: indeksni tasodifiy aralashtirish va $\lceil n/B \rceil$ ta paketga bo'lish.
 3. Har bir paket $\mathcal{B}$ uchun: oldinga yurish, $L_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \ell_k$, gradient $\mathbf{g} = \nabla_{\theta} L_{\mathcal{B}}$, yangilash $\theta \leftarrow \theta - \eta \mathbf{g}$.
 4. Validatsiya yo'qotishini kuzatish; yaxshilanish to'xtasa — to'xtatish.
- $B = n$ — to'liq (batch) gradient tushish, $B = 1$ — sof stoxastik. $\mathbf{g}$ to'liq gradientning siljimagan bahosi: $\mathbb{E}[\mathbf{g}] = \nabla L$.

	Perceptron	Logistik neyron
Chiqish	$1[z \geq 0] \in \{0, 1\}$	$\sigma(z) \in (0, 1)$ — ehtimollik
Yo'qotish	$\max(0, -\tilde{y}z)$	$\ln(1 + e^z) - yz$ (binary cross-entropy)
Yangilash	faqat xatoda: $\eta \tilde{y} \mathbf{x}$	har doim: $-\eta(p - y) \mathbf{x}$
Ajralmaydigan to'plam	yaqinlashmaydi	yagona optimumga intiladi (qavariq)
Ajraladigan to'plam	chekli qadamda to'xtaydi	$\ \mathbf{w}\ \rightarrow \infty$ (regulyarizatsiya kerak)

5.3. Bir yashirin qatlamli tarmoq

$X \in \mathbb{R}^{n \times d}$ (qatorlar — namunalar), $W_1 \in \mathbb{R}^{d \times m}$, $\mathbf{b}_1 \in \mathbb{R}^m$, $\mathbf{w}_2 \in \mathbb{R}^m$, $b_2 \in \mathbb{R}$:

$$Z_1 = XW_1 + \mathbf{1b}_1^T, \quad H = \varphi(Z_1), \quad \mathbf{z} = H\mathbf{w}_2 + b_2\mathbf{1}, \quad L = \frac{1}{n} \sum_k [\ln(1 + e^{z_k}) - y_k z_k].$$

Zanjir qoidasini chiqishdan kirishga qo'llab (to'liq isboti 2-ma'ruzada):

$$\begin{aligned} \mathbf{g}_2 &= \frac{1}{n} (\sigma(\mathbf{z}) - \mathbf{y}), & \nabla \mathbf{w}_2 &= H^T \mathbf{g}_2, & \frac{\partial L}{\partial b_2} &= \mathbf{1}^T \mathbf{g}_2, \\ G_1 &= (\mathbf{g}_2 \mathbf{w}_2^T) \odot \varphi'(Z_1), & \nabla W_1 &= X^T G_1, & \nabla \mathbf{b}_1 &= G_1^T \mathbf{1}. \end{aligned}$$

O'lchamlar tekshiruvi: $H^T \mathbf{g}_2$ — $m \times 1$, $X^T G_1$ — $d \times m$, ya'ni har bir gradient o'z parametri o'lchamida.

Misol 5.8 (Oldinga va orqaga yurish qo'lda). $\mathbf{x} = (1, 2)$, $y = 0$, ReLU yashirin qatlam, $W_1 = \begin{pmatrix} 1 & -1 \\ 0,5 & 1 \end{pmatrix}$, $\mathbf{b}_1 = (0; -0,5)$, $\mathbf{w}_2 = (0,5; -1)$, $b_2 = 0$ ($n = 1$).

Yechim. Oldinga. $Z_1 = (1 \cdot 1 + 2 \cdot 0,5 + 0; 1 \cdot (-1) + 2 \cdot 1 - 0,5) = (2; 0,5)$, $H = (2; 0,5)$ (ikkalasi musbat). $z = 2 \cdot 0,5 + 0,5 \cdot (-1) = 0,5$, $p = \sigma(0,5) \approx 0,6225$, $\ell = -\ln(1 - p) \approx 0,9741$.

Orqaga. $g_2 = p - y = 0,6225$; $\nabla \mathbf{w}_2 = g_2 \mathbf{H}^T = (1,2449; 0,3112)$, $\partial \ell / \partial b_2 = 0,6225$. $G_1 = g_2 \mathbf{w}_2^T \odot (1, 1) = (0,3112; -0,6225)$,

$$\nabla W_1 = \mathbf{x}^T G_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} (0,3112; -0,6225) = \begin{pmatrix} 0,3112 & -0,6225 \\ 0,6225 & -1,2449 \end{pmatrix}, \quad \nabla \mathbf{b}_1 = (0,3112; -0,6225).$$

$\eta = 0,5$ bilan bitta qadamdan keyin $z = -3,1776$, $p \approx 0,0400$, $\ell \approx 0,0408$. Barcha qiymatlar PyTorch autograd bilan mos keladi. Talqin: $y = 0$, lekin $p > 0,5$ — barcha gradientlar z ni kamaytiradigan yo'nalishda; ∇W_1 ning ikkinchi qatori birinchisidan ikki marta katta, chunki $x_2 = 2x_1$. $\triangleleft$

5.4. Boshlang'ich qiymatlar (initialization)

Simmetriya. Agar $W_1 = 0$ (yoki barcha ustunlari teng) bo'lsa, barcha yashirin neyronlar bir xil chiqish va bir xil gradient oladi, yangilangandan keyin ham teng qoladi: m ta neyron aslida bitta neyronga aylanadi. Shu sababli og'irliklar tasodifiy boshlanadi (siljishlar nol bo'lishi mumkin).

Dispersiya. $z_j = \sum_{i=1}^d w_{ij} a_i$, w_{ij} mustaqil, $\mathbb{E} w = 0$, $\text{Var } w = v$ bo'lsa, $\text{Var } z_j = d v \mathbb{E}[a_i^2]$. Signal qatlamdan qatlamga o'smasligi va so'nmasligi uchun $d v \mathbb{E}[a^2] \approx \text{Var } z$ kerak. $\tanh$ uchun (nol atrofiga $\varphi(z) \approx z$) *Xavier/Glorot*: $v = \frac{2}{d_{\text{in}} + d_{\text{out}}}$; ReLU uchun $\mathbb{E}[a^2] = \frac{1}{2} \text{Var } z$ (simmetrik z ning yarmi nolga aylanadi), shuning uchun *He* (Kaiming): $v = \frac{2}{d_{\text{in}}}$. PyTorch'da `nn.init.xavier_normal_`, `nn.init.kaiming_normal_`.

Teorema 5.9 (Universal approssimatsiya; Cybenko 1989, Hornik 1991, Leshno va boshq. 1993). φ uzluksiz va ko'phad bo'lmasin. U holda har qanday kompakt $K \subset \mathbb{R}^d$, uzluksiz $f : K \rightarrow \mathbb{R}$ va $\varepsilon > 0$ uchun shunday m va parametrlar topiladiki, $g(\mathbf{x}) = \sum_{j=1}^m v_j \varphi(\mathbf{w}_j^T \mathbf{x} + b_j)$ uchun $\sup_{\mathbf{x} \in K} |f(\mathbf{x}) - g(\mathbf{x})| < \varepsilon$.

Teorema *mavjudlikni* kafolatlaydi, lekin m qanchalik katta bo'lishini, parametrlarni gradient tushish topa olishini va modelning umumlashtirishini kafolatlamaydi. Amalda chuqur (ko'p qatlamli) tarmoqlar xuddi shu funksiyalarni ancha kam neyron bilan ifodalaydi.

Amaliy talqin: XOR tipidagi qarorlar

“Ikki omildan faqat bittasi bo'lsa xavf yuqori” tipidagi qoidalar (masalan, ikki dori alohida xavfsiz, lekin birgalikda xavfli; yoki aksincha) — XOR tuzilmasi. Bunday bog'liqlikni chiziqli skoring modeli ushlay olmaydi: $x_1 x_2$ o'zaro ta'sir belgisini qo'lda qo'shish yoki yashirin qatlam kerak. Chuqur o'rganishning afzalligi — bunday belgilarni ma'lumotdan avtomatik o'rganish.

6. Python'da amalga oshirish

1. Perceptron (NumPy). Algoritm 1 ning to'g'ridan-to'g'ri tatbiqi.

```
import numpy as np
X = np.array([[0,0],[0,1],[1,0],[1,1]], dtype=float)

def perceptron(X, y, eta=1.0, T=100):
    w, b = np.zeros(X.shape[1]), 0.0
    for t in range(1, T+1):
        err = 0
        for x, yk in zip(X, y):
            d = yk - int(w @ x + b >= 0) # y - y_hat
            if d != 0:
                w, b, err = w + eta*d*x, b + eta*d, err + 1
        if err == 0:
            return w, b, t, True
    return w, b, T, False

print(perceptron(X, np.array([0,0,0,1]))) # (array([2., 1.]), np.float64(-3.0), 6, True)
print(perceptron(X, np.array([0,1,1,0]))) # (array([-1., 0.]), np.float64(0.0), 100, False)
```

2. XOR uchun 2–4–1 tarmoq (qo'lda yozilgan gradient, tanh).

```
y = np.array([[0],[1],[1],[0]], dtype=float)
rng = np.random.default_rng(1)
W1 = rng.normal(0, 1, (2, 4)); b1 = np.zeros(4)
W2 = rng.normal(0, 1, (4, 1)); b2 = np.zeros(1)
eta, n = 0.5, len(X)
for s in range(2001):
    Z1 = X @ W1 + b1; H = np.tanh(Z1) # oldinga yurish
    z = H @ W2 + b2
    L = np.mean(np.logaddexp(0, z) - y*z) # barqaror BCE
    G2 = (1/(1+np.exp(-z)) - y) / n # dL/dz = (p - y)/n
    gW2, gb2 = H.T @ G2, G2.sum(0)
    G1 = (G2 @ W2.T) * (1 - H**2) # tanh' = 1 - tanh^2
    gW1, gb1 = X.T @ G1, G1.sum(0)
    W1 -= eta*gW1; b1 -= eta*gb1; W2 -= eta*gW2; b2 -= eta*gb2
    if s % 500 == 0: print(s, round(L, 4))
p = 1/(1+np.exp(-(np.tanh(X @ W1 + b1) @ W2 + b2)))
print(np.round(p.ravel(), 3))
# 0 0.6712 | 500 0.0114 | 1000 0.0051 | 1500 0.0033 | 2000 0.0024
# [0. 0.997 0.997 0.004]
```

Agar generator `default_rng(0)` bilan ishga tushirilsa, yo'qotish $\approx 0,349$ da to'xtab qoladi va bashorat (0,002; 0,499; 0,999; 0,501) bo'ladi — gradient tushish *lokal* usul, natija boshlang'ich qiymatga bog'liq. Yashirin neyronlar sonini 8 ga oshirish bu seed bilan ham muammoni hal qiladi (ortiqcha parametrlash optimallashtirishni osonlashtiradi). Qo'lda yozilgan $\nabla W_1, \nabla W_2$ PyTorch autograd natijasidan $\leq 1,6 \cdot 10^{-17}$ ga farq qiladi.

Faollashtirish funksiyalarini taqqoslash. Xuddi shu 2–4–1 tarmoq, $\eta = 0,5$, 2000 qadam, seed = 0, ..., 9 (`default_rng`), yashirin qatlamda turli φ :

φ	Yechildi	Mediana L	Izoh
sigmoid	10	0,0115	barqaror, lekin sekin ($\sigma' \leq 0,25$)
tanh	9	0,0024	seed 0 da plato ($L \approx 0,349$)
ReLU	5	0,1744	har bir muvaffaqiyatsiz holatda 2–3 neyron “o'lik”
Leaky ReLU	7	0,0021	$\alpha = 0,01$; gradient to'liq yo'qolmaydi

Xulosa: kichik tarmoqda ReLU ning nol gradientli sohasi xavfli; amalda bunga kengroq qatlam, He boshlang'ich qiymati, Leaky ReLU yoki Adam kabi optimizatorlar bilan qarshi turiladi. Tajriba natijasi seed ga bog'liq — bitta ishga tushirishdan xulosa chiqarmang.

3. Autograd bilan tekshirish (PyTorch) — 5.3-misol.

```
import torch
x = torch.tensor([1.0, 2.0]); y = torch.tensor(1.0)
w = torch.tensor([0.5, -0.3], requires_grad=True)
b = torch.tensor(0.1, requires_grad=True)
L = torch.nn.functional.binary_cross_entropy_with_logits(w @ x + b, y)
L.backward()
print(round(L.item(), 4), w.grad, b.grad) # 0.6931 tensor([-0.5000, -1.0000]) tensor(-0.5000)
```

`binary_cross_entropy_with_logits` sigmoid va BCE ni birlashtirib, aynan barqaror $\ln(1 + e^z) - yz$ shaklini hisoblaydi; shuning uchun modelning oxiriga alohida Sigmoid qo'yilmaydi.

7. Xulosa

- Neyron: $a = \varphi(\mathbf{w}^T \mathbf{x} + b)$; chiziqli qism giperyuza beradi, $\mathbf{w}$ — normal, $|z|/\|\mathbf{w}\|$ — masofa.
- Chiziqsizliksiz qatlamlar kompozitsiyasi bitta affin qatlamga teng — chuqurlikning foydasi φ dan.
- $\sigma' = \sigma(1 - \sigma) \leq \frac{1}{4}$, $\tanh = 2\sigma(2z) - 1$; sigmoid chuqur tarmoqda gradientni so'ndiradi, ReLU esa yo'q.
- Perceptron ajraluvchi to'plamda ko'pi bilan $(R/\gamma)^2$ ta xato bilan to'xtaydi (Novikoff); ajralmaydigan to'plamda to'xtamaydi.
- XOR chiziqli ajralmaydi; ikki yashirin ReLU neyroni uni aniq hisoblaydi.
- Sigmoid + binary cross-entropy: $\partial \ell / \partial z = p - y$; barqaror shakl $\ln(1 + e^z) - yz$.

7. Bir yashirin qatlam har qanday uzluksiz funksiyani approksimatsiya qila oladi, lekin bu o'qitish osonligini kafolatlamaydi.

8. Nazorat savollari

1. Sun'iy neyron qanday ta'riflanadi? Uning parametrlari nechta?
2. Siljish b ning geometrik ma'nosi nima?
3. Nima uchun $\mathbf{w}$ giperyuzaga normal?
4. Chiziqliksiz chuqur tarmoq nega bir qatlamga teng?
5. $\sigma' = \sigma(1 - \sigma)$ ni isbotlang.
6. Nima uchun $\sigma'(z) \leq 0,25$?
7. Gradientning so'nishi nima va u qaysi funksiyalarda kuchli?
8. "O'lik ReLU" nima? Leaky ReLU qanday yordam beradi?
9. Perceptron qoidasini yozing va uning mantiqini tushuntiring.
10. Novikoff teoremasining sharti va xulosasi nima?
11. Nega perceptron ajralmaydigan to'plamda to'xtamaydi?
12. XOR ning chiziqli ajralmasligini isbotlang.
13. Yashirin qatlam XOR ni qanday yechadi?
14. Binary cross-entropy ning barqaror shakli qanday?
15. Nega $\partial \ell / \partial z = p - y$?
16. Universal approksimatsiya teoremasi nimani kafolatlamaydi?

9. Mashqlar (javoblari bilan)

A guruh.

1. $\mathbf{x} = (2; -1)$, $\mathbf{w} = (0,5; 1)$, $b = 0,5$: z va sigmoid, tanh, ReLU chiqishlari. *Javob:* $z = 0,5; 0,6225; 0,4621; 0,5$.
2. $\sigma'(0)$, $\tanh'(0)$, $\sigma'(2)$ ni hisoblang. *Javob:* $0,25; 1; \sigma(2) \approx 0,8808$, $\sigma'(2) \approx 0,1050$.
3. $(1, 1)$ nuqtadan $3x_1 + 4x_2 - 2 = 0$ to'g'ri chiziqqacha masofa. *Javob:* $|3 + 4 - 2|/5 = 1$.
4. Leaky ReLU ($\alpha = 0,01$) va uning hosilasi $z = -3$ da. *Javob:* $-0,03; 0,01$.
5. OR uchun perceptron ($\eta = 1$, $\mathbf{w}_0 = \mathbf{0}$, $b_0 = 0$, 3.1-misoldagi tartib). *Javob:* 4-epoxada xatosiz; $\mathbf{w} = (1, 1)$, $b = -1$ (xatolar: $2, 2, 1, 0$).
6. NAND ni hisoblaydigan perceptron og'irliklarini tanlang. *Javob:* masalan $\mathbf{w} = (-1, -1)$, $b = 1,5$: $z = (1,5; 0,5; 0,5; -0,5)$.
7. $z = 3$, $y = 0$ uchun sigmoid chiqishli neyronda binary cross-entropy va MSE ($\frac{1}{2}(p - y)^2$) bo'yicha $\partial \ell / \partial z$ ni solishtiring. *Javob:* $p \approx 0,9526$; $0,9526$ va $\approx 0,0430$ (22 marta kichik).
8. $d = 3$ kirishli, $m = 5$ neyronli qatlam: parametrlar soni va $n = 32$ paket uchun Z o'lchami. *Javob:* $5 \cdot 4 = 20$; 32×5 .

B guruh.

9. $\tanh'(z) = 1 - \tanh^2 z$ ni Teorema 2.1(i),(iv) dan chiqaring. *Ko'rsatma:* $\tanh' = 4\sigma'(2z) = 4s(1 - s)$, $s = \sigma(2z) = (1 + \tanh z)/2$.
10. Softplus hosilasi sigmoid ekanini isbotlang; $\lim_{z \rightarrow \pm\infty} (\text{softplus}(z) - \text{ReLU}(z)) = 0$ ni ko'rsating.
11. $\mathbf{x} = (1; -1)$, $y = 0$, $\mathbf{w} = \mathbf{0}$, $b = 0$, $\eta = 1$: bitta gradient qadami va yangi yo'qotish. *Javob:* $\nabla_{\mathbf{w}} \ell = (0,5; -0,5)$, $\partial \ell / \partial b = 0,5$; $\mathbf{w} = (-0,5; 0,5)$, $b = -0,5$, $z = -1,5$, $p \approx 0,1824$, $\ell \approx 0,2014$ (avval $\ln 2$).
12. $784 - 128 - 64 - 10$ to'liq bog'langan tarmoq parametrlari soni. *Javob:* $100 \cdot 480 + 8 \cdot 256 + 650 = 109 \cdot 386$.
13. Bitta yashirin neyronli tarmoq XOR ni yecha olmasligini isbotlang. *Ko'rsatma:* chiqish $u = \mathbf{w}_1^T \mathbf{x} + b_1$ ning monoton funksiyasi.
14. $\tilde{z} = x_1 + x_2 - 2x_1x_2 - 0,5$ bilan XOR bitta neyronda yechilishini tekshiring. *Javob:* $\tilde{z} = (-0,5; 0,5; 0,5; -0,5)$ — ajraladi (qo'shimcha belgi x_1x_2).

C guruh.

15. AND uchun $\tilde{\mathbf{u}} \propto (2, 2, -3)$ bilan Novikoff chegarasini hisoblang va 3.1-misol bilan solishtiring. *Javob:* $R = \sqrt{3}$, $\gamma = 1/\sqrt{17}$, $(R/\gamma)^2 = 51$; haqiqatda 11 ta yangilanish.
16. L ta chiziqli qatlamli tarmoq ($\varphi(z) = z$) uchun ekvivalent $W, \mathbf{b}$ ni yozing va isbotlang. *Javob:* $W = W_L \cdots W_1$, $\mathbf{b} = \sum_{i=1}^L W_L \cdots W_{i+1} \mathbf{b}_i$.
17. Agar barcha namunalarda $z_j < 0$ bo'lsa, ReLU neyroni j ning kiruvchi og'irliklari gradienti nol ekanini ko'rsating ("o'lik ReLU"). *Ko'rsatma:* G_1 ning j -ustuni $\varphi'(Z_1)$ ning nol ustuniga ko'paytiriladi.
18. $2-m-1$ tarmoqni `default_rng(0)` bilan $m = 2, 4, 8$ uchun o'qiting; yakuniy yo'qotishlarni solishtiring va natijani lokal minimum/plato tushunchalari bilan izohlang.

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